MCTE

CONCEPT NOTES

AI Modelling, Big Data Analytics & Biometrics

Ten Concept Notes covering State-of-the-Art, Identified Gaps, Objectives, Novelty, Proposed Architecture and Conclusions

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1. Development of AI-based Noise Removal Tool for Radio Intercepts

1. Background

Radio intercept analysis is a cornerstone of Signals Intelligence (SIGINT). Modern battlefields are saturated with electromagnetic emissions across HF, VHF, UHF and microwave bands, yet adversarial communication systems deliberately embed signals in high-noise environments, use frequency hopping and employ spread-spectrum techniques to evade detection. State-of-the-art noise suppression has evolved from classical Wiener filtering and spectral subtraction towards deep learning architectures. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have demonstrated superior denoising on speech signals. More recently, transformer-based models such as Denoising Diffusion Probabilistic Models (DDPMs) and attention-gated U-Nets achieve near-human intelligibility scores on noisy audio. Foundation models pre-trained on large corpora (e.g., Meta's EnCodec, OpenAI Whisper) provide strong priors for restoration tasks. In military contexts, Qualcomm, DARPA's GARD program and various NATO STANAG research groups have published results on adversarial-robust denoising. However, most commercial systems are optimised for civil telephony rather than tactical wideband intercepts, encrypted bursts, or co-channel interference patterns unique to military operations.

2. Identified Gaps

- Existing commercial denoising tools are trained on speech corpora (VCTK, LibriSpeech) and fail on military waveforms (HAVE QUICK, Link-16, SINCGARS).
- No open-source tool handles simultaneous co-channel jamming and thermal noise in real-time on edge hardware.
- Adversarial robustness of denoising models against deliberate signal poisoning or spoofing is largely unaddressed.
- Current solutions lack indigenous development, creating supply-chain vulnerability for Indian defence systems.
- Real-time operation under SWaP (Size, Weight and Power) constraints of forward-deployed units is unsupported by GPU-heavy models.

3. Objectives

- Design a deep neural network pipeline for broadband noise removal optimised for military radio waveforms.
- Achieve real-time processing on embedded hardware (NVIDIA Jetson / custom FPGA).
- Build a labelled dataset of synthetic and live military-grade noisy radio intercepts for supervised training.
- Incorporate adversarial training regimes to resist deliberate noise injection attacks.

4. Novelty in the Proposed Solution

The proposed solution introduces a Hybrid Spectro-Temporal Attention Network (HSTAN) that combines a mel-spectrogram encoder with a bidirectional temporal attention decoder, specifically fine-tuned on synthesised military waveform noise profiles (jamming, multi-path fading, burst interference). A novel data-augmentation pipeline generates adversarial noise samples using Generative Adversarial Networks (GANs) to harden the model. Unlike commercial solutions, HSTAN operates in a two-stage cascade: a coarse spectral mask prediction followed by a fine-grained residual waveform synthesis, enabling deployment on edge devices without cloud dependency. The architecture is the first of its kind to incorporate Indian military radio protocol awareness as an inductive bias in the network design.

5. Proposed Architecture

The system comprises five modules: (1) Wideband SDR Front-End that digitises RF signals from 1 MHz to 6 GHz and outputs IQ samples; (2) Feature Extraction Module performing short-time Fourier transform (STFT) and mel-filterbank computation; (3) HSTAN Core consisting of a CNN encoder (5 layers, 512 channels), bidirectional LSTM temporal modeller, and multi-head cross-attention decoder generating a time-frequency mask; (4) Waveform Reconstruction Module applying the mask and performing inverse STFT with Griffin-Lim phase recovery; and (5) Edge Deployment Layer compressing the model via quantisation-aware training (INT8) for FPGA/Jetson deployment. A continuous online learning loop adapts the model to new noise signatures encountered in the field.

6. Conclusion

Adversarial communication environments demand noise-robust intercept analysis tools that operate indigenously on Indian Army platforms. The proposed HSTAN-based tool directly addresses this critical gap by providing a real-time, edge-deployable, military waveform-aware denoising capability. Its development will significantly enhance SIGINT operational effectiveness, reduce reliance on foreign technology and contribute to the broader vision of self-reliant defence AI under Atmanirbhar Bharat.

2. AI Module for Real-Time Video Analytics and Anomaly Detection

1. Background

The proliferation of surveillance cameras in forward operating bases, border posts, ammunition depots and critical infrastructure has created an overwhelming volume of video data that human operators cannot monitor effectively. Real-time video analytics has advanced dramatically with the maturation of deep learning. Object detection architectures such as YOLOv8, RT-DETR and DINO achieve mean Average Precision (mAP) exceeding 60% on COCO benchmarks at over 100 frames per second. Multi-object tracking algorithms (ByteTrack, BoT-SORT) enable persistent identity maintenance across occlusions. Anomaly detection in video, however, remains more challenging: methods range from reconstruction-based autoencoders (ConvLSTM-AE), prediction-based models (PredNet), to self-supervised contrastive methods (STOC). Recent vision-language models (CLIP, InternVideo) enable zero-shot anomaly recognition through natural language descriptions of normal behaviour. Military-specific challenges include camouflage detection, low-light operation, thermal-visible fusion and resilience to adversarial patches placed on objects by adversaries to blind AI cameras.

2. Identified Gaps

- Commercial CCTV analytics products are not calibrated for military scenes (camouflage, thermal, night-vision, IR sensors).
- Existing anomaly detection systems have high false-positive rates in dynamic outdoor military environments (wind, animals, foliage movement).
- No integrated solution exists that fuses visible, thermal and radar micro-Doppler data for multi-modal anomaly detection.
- Real-time alerting with human-readable natural language explanations is absent in current military systems.
- Adversarial robustness against patch attacks or scene manipulation by sophisticated adversaries is not addressed.

3. Objectives

- Develop a multi-sensor real-time video analytics engine.
- Achieve anomaly detection with acceptable precision and recall in military outdoor environments.
- Integrate multi-modal sensor fusion for holistic scene understanding.
- Provide explainable, natural language anomaly alerts to operators regarding event occurrence.

4. Novelty in the Proposed Solution

The proposed AI module introduces a Military Scene-Aware Transformer (MSAT) that fuses heterogeneous sensor streams using cross-modal attention. The novelty lies in a two-branch architecture: a spatial branch for object/activity recognition using fine-tuned DINO-v2, and a temporal branch capturing motion trajectories via graph neural network (GNN)-based trajectory modelling. Anomalies are detected by comparing real-time scene graphs against a dynamically updated Normal Behaviour Knowledge Base (NBKB) populated through unsupervised scene analysis during quiet periods. A vision-language module (based on LLaVA-style architecture) generates natural language explanations of detected anomalies, enabling non-expert operators to act rapidly. The system is hardened against adversarial patches through certified randomised smoothing at inference time.

5. Proposed Architecture

Architecture encompasses: (1) Multi-Sensor Ingestion Layer, thermal cameras and radar data feeds; (2) Preprocessing Pipeline for frame synchronisation, calibration and histogram equalisation; (3) MSAT Processing Core with spatial encoder (DINO-v2 backbone), temporal encoder (GNN + Transformer), and cross-modal fusion block; (4) Anomaly Scoring Engine comparing scene embeddings against NBKB using cosine similarity and isolation forest outlier scoring; (5) Alert Generation Module with LLaVA-based captioning and priority ranking; (6) Command Dashboard providing a GIS-integrated map view, video thumbnails and operator acknowledgment workflow; and (7) Edge GPU Cluster (NVIDIA A2/L4 per node) for distributed processing.

6. Conclusion

As the Indian Army deploys more sensors across its extended perimeter, the need for intelligent automated surveillance is urgent. The proposed MSAT-based real-time video analytics module will transform passive camera networks into an active, AI-driven early warning system, dramatically reducing response time to intrusions, sabotage and other anomalous events, while lowering operator cognitive burden through explainable AI alerts.

3. Development of Indigenous LLM for Indian Army with Advanced Reasoning and Multimodal Capability

1. Background

Large Language Models (LLMs) have redefined the frontier of artificial intelligence. Models such as GPT-4o, Gemini 1.5 Pro, Llama 3.1 and Mistral demonstrate emergent reasoning, multi-step planning and multimodal understanding across text, images, audio and documents. In the defence domain, NATO nations are actively developing sovereign LLMs: the UK's Dstl is evaluating LLMs for intelligence analysis; the US DoD's Task Force Lima is integrating LLMs into C2 workflows; and France's DGA is funding LLM research. India has made progress with IndicLLM Suite, Krutrim, and Sarvam AI's models, but none are tailored for defence-grade security, military terminolgy, operational doctrine, or classified data handling. An indigenous defence LLM must be deployable on air-gapped networks, support Indic languages, possess multimodal reasoning over maps/imagery/signals and adhere to strict data governance, making it fundamentally distinct from commercial offerings.

2. Identified Gaps

- No existing Indian LLM is trained on military doctrines, field manuals, SIGINT/IMINT reports, or operational after-action reviews.
- Commercial LLMs cannot be deployed on air-gapped, classified networks due to cloud dependency and foreign data residency.
- Support for Indic languages in military contexts is absent in current models.
- Multimodal reasoning over tactical maps, satellite imagery and radio intercept text simultaneously is unsupported.
- Hallucination rates in mission-critical reasoning tasks remain unacceptably high in general-purpose LLMs.

3. Objectives

- Pre-train indigenous LLM on curated military and open-source corpora with robust data governance.
- Fine-tune the model using Reinforcement Learning from Human Feedback (RLHF) on military reasoning tasks.
- Develop multimodal extensions for processing tactical maps, satellite imagery and structured signal data.
- Enable deployment on classified, air-gapped, on-premise GPU clusters without internet dependency.

4. Novelty in the Proposed Solution

The proposed LLM, tentatively named SAINIK-LLM (Secure AI for iNdigenous Intelligence and Knowledge), introduces three novel elements: (i) a Military Domain-Adaptive Pre-training (MDAP) corpus combining declassified doctrines, field manuals, OSINT and synthetic military data generated under expert supervision; (ii) a Hierarchical Multimodal Encoder that unifies text, tactical map features (extracted via specialised map-parsing CNNs) and satellite image embeddings within a shared latent space; and (iii) a Constitutional AI-inspired safety layer that enforces operational security (OPSEC) constraints, preventing the model from revealing classified reasoning pathways even to authenticated users without need-to-know clearance.

5. Proposed Architecture

The system is structured as: (1) Data Pipeline with classified document ingestion, OCR and de-classification tagging; (2) Pre-training Infrastructure using a 512-GPU cluster with ZeRO-3 sharding and mixed-precision training; (3) SAINIK-LLM Core (transformer decoder, 70B params, grouped-query attention, RoPE embeddings); (4) Multimodal Fusion Module with image and map encoders projecting into text embedding space via cross-attention adapters; (5) RAG Layer connecting to a secure military knowledge graph and vector database; (6) RLHF Pipeline with military SME annotators for preference data collection; and (7) Secure Inference Gateway enforcing role-based access control, audit logging and on-premise hosting.

6. Conclusion

An indigenous, air-gapped LLM for the Indian Army is a strategic imperative. SAINIK-LLM will serve as the cognitive backbone for next-generation C2 systems, intelligence analysis, logistics optimisation and training simulation—reducing dependence on foreign AI platforms and ensuring that India's military intellectual capital remains sovereign, secure and operationally superior.

4. Development of AI-Based Language Translation Tool for Signal Intelligence Units

1. Background

SIGINT units routinely intercept communications in dozens of languages, including rare dialects, coded language, military jargon and rapidly evolving slang. The state of the art in machine translation has been transformed by neural machine translation (NMT) systems: Google Translate and DeepL, built on Transformer architectures, achieve near-human BLEU scores for major language pairs. Multilingual models such as NLLB-200 (No Language Left Behind) support 200 languages, while Whisper-large-v3 enables multilingual automatic speech recognition (ASR) across 99 languages. However, these systems are designed for civilian use and fail on military-specific vocabulary, deliberate obfuscation, code-words and dialectal variations used in operational communication. Projects like DARPA BOLT (Broad Operational Language Translation) and NATO's Multi-Language Processing Working Group have produced domain-adapted systems but none are available to India as off-the-shelf sovereign solutions.

2. Identified Gaps

- Existing NMT systems lack military vocabulary, code-words and tactical jargon across operationally relevant languages (Mandarin, Pashto, Baloch, Urdu dialects, Tibetan).
- Real-time speech-to-speech translation with latency below 1 second for SIGINT operations is not available indigenously.
- Obfuscation detection — identifying deliberate use of code-words, euphemisms, or steganographic language — is absent.
- Offline, air-gapped deployment capability is not supported by cloud-dependent commercial translation services.
- Speaker diarisation and multi-speaker translation in radio intercepts (multiple simultaneous speakers) is unsolved.

3. Objectives

- Develop a multilingual NMT system covering 5 priority languages with military-domain fine-tuning.
- Achieve real-time speech-to-text-to-translation pipeline with end-to-end latency.
- Integrate an obfuscation and code-word detection module alerting analysts to intentionally obscured communications.
- Implement multi-speaker diarisation for accurate attribution in multi-participant intercepts.

4. Novelty in the Proposed Solution

The proposed tool, named VAANI (Voice and Analytics for Armed Forces National Intelligence), introduces a novel Military-Adapted Multilingual Transformer (MAMT) fine-tuned on a custom corpus of military communications, intercepted open-source broadcasts and synthetically generated tactical conversations across 15 languages. VAANI uniquely incorporates a Code-Word Inference Engine (CWIE) that learns latent semantic anomalies in communication patterns to flag potential obfuscated content. A speaker-independent diarisation module using self-supervised speech representations (HuBERT) enables accurate multi-speaker separation in noisy channel conditions.

5. Proposed Architecture

VAANI comprises: (1) RF-to-Text Frontend with SDR interface, noise removal (integrated with the Noise Removal Tool), and Whisper-based ASR; (2) MAMT Translation Engine with language identification, multi-language NMT (NLLB-200 fine-tuned), and back-translation confidence scoring; (3) CWIE with anomaly scoring against expected semantic distributions; (4) Speaker Diarisation Module (HuBERT + clustering); (5) Analyst Workstation with a rich UI showing side-by-side source/translation, confidence scores, flagged terms and audio playback; (6) Knowledge Base of military terminology, code-word dictionaries updated through analyst feedback; and (7) Secure Offline Runtime packaged as a containerised (Docker) appliance for field deployment.

6. Conclusion

Language remains the most critical barrier in SIGINT analysis. VAANI will equip Indian Army SIGINT units with an indigenous, offline-capable, militarily-aware language translation capability that goes beyond mere word-for-word conversion to deliver actionable intelligence—detecting obfuscation, attributing speakers and translating with contextual military precision across operationally critical languages.

5. Development of AI-Based Deep Fake Generator and Detector for Audio, Text and Video

1. Background

Synthetic media — commonly termed deepfakes — represents one of the most disruptive dual-use AI capabilities of the current decade. Generation technology has advanced from early GAN-based face-swap methods (DeepFaceLab, FaceSwap) to diffusion model-based full-scene synthesis (Stable Diffusion, Sora) and neural voice cloning (VALL-E, ElevenLabs). Text generation by LLMs (GPT-4, Claude) produces indistinguishable disinformation at scale. In the military domain, adversaries can generate fake surrender orders, fabricated intelligence reports, simulated battlefield imagery, or synthetic voice recordings of commanders to cause confusion, erosion of trust and OODA loop disruption. Detection systems have also advanced: FaceForensics++, DeepFake-o-meter, and Microsoft's Video Authenticator use forensic artefacts, GAN fingerprints, and biological signal analysis (rPPG) to identify synthetic content. However, the arms race between generators and detectors continues to accelerate, and no sovereign Indian capability exists for either offensive simulation or defensive detection.

2. Identified Gaps

- No indigenous capability exists for generating high-fidelity deepfakes for training detection models, PSYOPS simulation, or red-teaming.
- State-of-the-art detectors achieve only 70-80% accuracy on cross-dataset evaluations, demonstrating poor generalisation.
- Multi-modal deepfake detection (audio-visual consistency, text-image alignment) is not addressed by current single-modality detectors.
- Real-time detection capability for live video streams in tactical communication channels is absent.
- Provenance tracking and digital watermarking of authentic military media to distinguish from synthetics is not standardised.

3. Objectives

- Develop a controlled deepfake generation platform for training datasets, simulation exercises and red-team evaluation.
- Build a multi-modal deepfake detection system across audio, video and text modalities.
- Achieve real-time detection suitable for live video stream monitoring.
- Implement invisible digital watermarking for authentic Indian Army multimedia content.

4. Novelty in the Proposed Solution

The proposed dual-use platform, PRATYAKSH (Provenance, Reality Authentication and Truth Yielding AI for Krishna-arjuna Synthetic Hybridisation), innovates on two fronts. The Generator Module uses a conditional latent diffusion model with identity-preserving adapters, controllable lip-sync and neural voice cloning, specifically tuned to produce adversary-style synthetic media for training and simulation. The Detector Module introduces a Cross-Modal Consistency Network (CMCN) that simultaneously analyses video frames, associated audio and any accompanying text, detecting inconsistencies in biological signals (blink rate, micro-expressions, phoneme-viseme alignment) and semantic coherence. PRATYAKSH is the first system to embed steganographic watermarks invisible to the human eye into authentic content at the point of capture using a camera-level firmware module.

5. Proposed Architecture

The platform consists of: (1) Deepfake Generator Suite (latent diffusion backbone, LoRA-based identity adaptation, TTS/voice clone module, text LLM for disinformation text generation); (2) CMCN Detection Engine (video branch: EfficientNet + temporal attention; audio branch: wav2vec 2.0 feature extraction; text branch: RoBERTa classifier; fusion: cross-modal transformer); (3) Real-Time Processing Pipeline with NVENC hardware decoding and GPU inference at 30+ fps; (4) Watermarking Service embedding invisible robust watermarks using HiDDeN-style steganography at camera firmware; (5) Forensic Attribution Engine classifying detected synthetics by generative model family using model fingerprinting; and (6) Analyst Console showing heatmaps of manipulated regions, confidence scores and provenance certificates.

6. Conclusion

The deepfake threat to military command integrity, PSYOPS resilience and intelligence authenticity is rapidly escalating. PRATYAKSH provides the Indian Army with a sovereign dual-use capability: a controlled generation environment for training and simulation, paired with a state-of-the-art multi-modal detection system. Together, these tools will underpin the Army's information warfare readiness and ensure that its communication infrastructure remains trustworthy in the face of adversarial synthetic media.

6. Change Detection in Satellite Imagery for Military Intelligence Analysis

1. Background

Satellite imagery change detection is a foundational task in geospatial intelligence (GEOINT). With the commercialisation of small satellites, revisit rates have fallen to hours (Planet Labs: 24-hr global revisit; Maxar: sub-daily tasking), generating petabytes of imagery requiring automated analysis. Early change detection relied on image differencing and PCA; modern methods exploit deep learning. Siamese neural networks (Siamese-CD), attention-based transformers (ChangeFormer, BIT), and diffusion model-based detectors now achieve over 90% F1 on public benchmarks (LEVIR-CD, WHU-CD). Multi-temporal, multi-spectral, and SAR (Synthetic Aperture Radar) fusion extends detection capability to all-weather, all-time conditions. Military applications include monitoring adversary infrastructure construction, detecting equipment repositioning, tracking supply logistics and identifying camouflaged assets. Nations including the USA (NGA), China (CASC), and Israel (ImageSat) have mature GEOINT AI pipelines; India's ISRO and NTrO have foundational capabilities but lack integrated AI-driven change detection tailored to military intelligence workflows.

2. Identified Gaps

- Current open-source change detection models are trained on civilian datasets and fail to detect military-specific changes (camouflage nets, hardened shelters, dispersal patterns).
- Multi-modal change detection fusing optical, SAR and hyperspectral imagery simultaneously is not available as an integrated solution.
- Automated false alarm rejection for natural changes (seasonal vegetation, flooding, agricultural cycles) in military surveillance zones is immature.
- Change prioritisation and intelligence report generation are manual, creating analyst bottlenecks.
- Real-time or near-real-time ingestion of commercial satellite feeds for tactical decision-making is not implemented indigenously.

3. Objectives

- Develop a deep learning change detection pipeline optimised for military GEOINT covering optical, SAR and hyperspectral data.
- Automate natural change suppression to isolate human-activity-related changes of military significance.
- Generate structured intelligence reports from detected changes using LLM-based narrative synthesis.
- Enable near-real-time processing of multi-satellite feeds from image capture to alert.

4. Novelty in the Proposed Solution

The proposed system, DRISHTI (Deep-learning Remote Intelligence for Surveillance, Hyperspectral and Temporal Interpretation), introduces a Tri-Modal Change Transformer (TMCT) that processes optical, SAR and hyperspectral image stacks through modality-specific encoders before fusing via a cross-modal attention mechanism. A novel Military Change Ontology (MCO) comprising 150+ military-specific change classes (vehicle dispersal, runway extension, missile erector deployment, etc.) guides the classification head. Natural change suppression employs a phenological model of seasonal vegetation and weather patterns as a prior to mask irrelevant changes. An LLM-based report generator transforms structured change detections into formatted intelligence assessments following Indian Army intelligence reporting standards.

5. Proposed Architecture

DRISHTI is organised as: (1) Satellite Data Ingestion Hub with APIs to ISRO Cartosat, RISAT, commercial SAR, and optical providers; (2) Pre-processing Pipeline for orthorectification, atmospheric correction, co-registration and pan-sharpening; (3) TMCT Core with independent SAR encoder (complex-valued CNN), optical encoder (Swin Transformer) and hyperspectral encoder (spectral attention network), fused via cross-attention; (4) MCO-guided Classification Head producing change type, confidence and geographic coordinates; (5) Natural Change Filter using phenological and meteorological suppression masks; (6) LLM Report Generator producing intelligence assessments; and (7) GEOINT Dashboard with interactive map overlays, time-series animation and alert dissemination to C2 systems.

6. Conclusion

Continuous monitoring of adversary activity across land borders, LOC and LAC demands an automated, AI-driven change detection capability that no human analyst team can match in speed or coverage. DRISHTI will provide the Indian Army with a sovereign, integrated GEOINT AI pipeline capable of detecting militarily significant changes in near-real-time, compressing the intelligence cycle from days to minutes and enabling proactive rather than reactive decision-making.

7. Development of Digital Twins for Military Operations Planning and Infrastructure

1. Background

A digital twin is a dynamic, physics-informed virtual replica of a physical entity that is continuously updated through real-world sensor data. Defence applications of digital twins have accelerated significantly: the US Army's Synthetic Training Environment (STE) integrates One World Terrain (OWT) digital twins of global terrain; the UK MoD's Project MARSHALL uses digital twins for aircraft maintenance; and NATO's Digital Twin of the Battlespace concept envisions real-time virtual mirrors of operational areas. Enabling technologies include BIM (Building Information Modelling), GIS-fused 3D terrain modelling, IoT sensor networks, physics simulation engines (Unreal Engine 5, NVIDIA Omniverse) and AI-driven predictive analytics. In the Indian context, digital twins could transform planning for mountain warfare, riverine operations, joint logistics and infrastructure survivability assessment. ISRO and NIC have initiated geospatial digital twin frameworks, but military-specific operational planning twins with real-time sensor fusion and AI-driven COA (Course of Action) analysis do not yet exist indigenously.

2. Identified Gaps

- No indigenous digital twin platform exists for military operational planning covering terrain, force disposition and logistics simultaneously.
- Real-time update of digital twins from heterogeneous sensor networks (drones, satellites, ground sensors, weather stations) is not integrated.
- AI-driven Course of Action (COA) analysis and wargaming within digital twin environments is absent.
- Infrastructure survivability assessment under kinetic and non-kinetic threats using physics-based simulation is not automated.
- Interoperability with existing Indian Army C2 systems (BMS, CIDSS, ACCCS) for real-time data exchange is not established.

3. Objectives

- Construct high-fidelity 3D digital twins of key operational areas using multi-source geospatial data (LiDAR, SAR, optical).
- Integrate real-time sensor data streams to maintain live synchronisation between physical and virtual environments.
- Develop AI-driven COA planning and automated wargaming modules for mission rehearsal and decision support.
- Enable infrastructure survivability analysis through physics-based simulation of damage and degradation scenarios.

4. Novelty in the Proposed Solution

The proposed platform, KSHITIJ (Knowledge-Synchronised High-fidelity Intelligence Twin for Joint Indian wargaming), introduces the concept of a Living Operational Digital Twin (LODT) that is continuously synchronised with ground truth at cadences from minutes (drone feeds) to milliseconds (sensor telemetry). KSHITIJ uniquely embeds an AI Wargaming Engine that uses Monte Carlo Tree Search (MCTS) enhanced with a military tactics neural network to generate and evaluate COAs within the digital twin. A Predictive Infrastructure Integrity Module (PIIM) applies finite element analysis (FEA) accelerated by physics-informed neural networks (PINNs) to assess bridge load capacities, runway structural integrity and bunker survivability under specified threat scenarios in near-real time. The system is the first Indian defence digital twin to natively integrate with BMS-NG data streams.

5. Proposed Architecture

KSHITIJ is built on: (1) Geospatial Foundation Layer with 1-metre DEM, building footprints, road networks and vegetation from ISRO/NIC; (2) Real-Time Sensor Fusion Hub ingesting drone video, satellite updates, IoT ground sensors, and weather data; (3) 3D Rendering and Physics Engine (Unreal Engine 5 + NVIDIA Omniverse for collaborative viewing); (4) AI Wargaming Module (MCTS + tactics neural network, blue/red force modelling); (5) PIIM for infrastructure survivability using PINNs; (6) C2 Integration Layer with BMS-NG, CIDSS, and ACCCS adapters; (7) Collaborative Planning Interface allowing multi-user, multi-site simultaneous planning sessions with role-based access; and (8) Classified Hosting on MoD-approved private cloud infrastructure.

6. Conclusion

Military operations planning, particularly in India's geographically diverse operational theatres, demands tools that can synthesise vast amounts of terrain, weather, logistic and threat data into actionable situational awareness. KSHITIJ will provide commanders with an immersive, AI-augmented digital twin environment that transforms the planning cycle, enables risk-free rehearsal of complex operations and provides predictive infrastructure survivability assessment — constituting a decisive leap in India's operational planning capability.

8. AI Tool for Vulnerability Analysis and Validation of AI Models

1. Background

As AI systems are embedded into increasingly critical military decision-making processes — from autonomous UAVs to AI-assisted fire control — the security and reliability of these AI models becomes a national security concern. Adversarial machine learning has demonstrated that AI models can be fooled by imperceptible input perturbations (adversarial examples), poisoned training data, model extraction attacks and membership inference attacks. Foundational works by Goodfellow (FGSM), Carlini-Wagner (C&W attack), and subsequent research have established a broad taxonomy of attacks. Defences including adversarial training, certified robustness (randomised smoothing), input preprocessing, and ensemble methods have been proposed, but no comprehensive automated tool for vulnerability assessment of military AI systems exists. The NIST AI Risk Management Framework, EU AI Act, and US DoD AI Ethics Principles all mandate AI validation; however, standardised tools for military AI validation — covering tactical AI, computer vision systems, and LLMs — remain nascent. India lacks a sovereign AI vulnerability assessment tool aligned with its own military AI deployment context.

2. Identified Gaps

- No sovereign Indian tool exists for systematic adversarial robustness testing of military AI models across modalities (vision, NLP, speech).
- Existing toolkits (IBM ART, Cleverhans, Foolbox) are research-grade and lack military-specific threat models and reporting formats.
- Validation of LLM-based military AI for hallucination, OPSEC leakage and prompt injection vulnerabilities is not standardised.
- Automated red-teaming pipelines that simulate sophisticated adversarial actors against deployed military AI are absent.
- Compliance reporting aligned with Indian defence AI governance frameworks is not supported by any existing tool.

3. Objectives

- Develop a comprehensive AI vulnerability assessment platform covering adversarial robustness, data poisoning, model extraction and LLM-specific threats.
- Automate adversarial red-teaming pipelines simulating nation-state-level adversarial capabilities.
- Provide a continuous monitoring capability for detecting model degradation and emerging vulnerabilities post-deployment.
- Validate AI models across all modalities deployed in Indian Army systems (vision, NLP, speech, multimodal).

4. Novelty in the Proposed Solution

The proposed platform, KAVACH (Knowledge-driven Adversarial Vulnerability Assessment and Certification Hub), is the first Indian sovereign tool designed specifically for military AI validation. KAVACH introduces an Adaptive Threat Emulation Engine (ATEE) that constructs threat scenarios based on an adversary capability model derived from open-source intelligence on nation-state AI attack capabilities. Unlike generic toolkits, KAVACH includes a Military AI Threat Taxonomy covering 200+ attack vectors specific to defence AI systems (sensor spoofing attacks on UAV vision, prompt injection against LLM-assisted C2 systems, acoustic attacks on voice biometrics). A novel Certification Scoring Framework (CSF) provides a quantitative security score for each AI module, enabling procurement and deployment decisions based on verified robustness thresholds.

5. Proposed Architecture

KAVACH consists of: (1) Model Ingestion Interface supporting PyTorch, TensorFlow, ONNX, and API-based black-box models; (2) Adversarial Attack Library with 200+ pre-implemented attacks across white-box, grey-box, and black-box threat models; (3) ATEE constructing realistic multi-step attack campaigns mimicking nation-state actors; (4) LLM Evaluation Module testing for hallucination rates, OPSEC leakage, prompt injection and jailbreak susceptibility; (5) Data Poisoning Simulator for training pipeline integrity assessment; (6) Reporting Engine generating structured vulnerability reports with CVSS-like severity scoring, remediation guidance, and compliance mapping; (7) Continuous Monitoring Agent deployed alongside production AI systems to detect distribution shifts and performance degradation in real time.

6. Conclusion

Trust in AI systems for military decision-making cannot be assumed — it must be earned through rigorous, systematic testing. KAVACH provides the Indian Army with a sovereign, military-contextualised AI vulnerability assessment capability that ensures every AI system deployed in operational roles has been stress-tested against realistic adversarial threats. This capability is foundational to responsible AI adoption in Indian defence and will support the development of an indigenous AI governance and certification ecosystem.

9. Image-Based Person Identification System for Military Access Control

1. Background

Person identification through imaging — encompassing face recognition, gait analysis, and body silhouette matching — has matured into a highly accurate and operationally deployable technology. State-of-the-art face recognition systems (ArcFace, AdaFace, FaceTransformer) achieve over 99.8% accuracy on LFW and IJB-C benchmarks. Gait recognition systems (GaitGraph, OpenGait) enable person identification at distances of 50–500 metres, even when faces are obscured. Whole-body re-identification (Person Re-ID) using BoT, TransReID, and SOLIDER models supports tracking individuals across multiple non-overlapping camera views. In military access control, these capabilities can significantly augment or replace traditional card/PIN-based systems, providing passive, contactless and continuous authentication. The Indian Army currently relies on physical identity cards, barcode scanners and limited biometric readers. There is no integrated, AI-driven image-based identification system that combines multi-distance recognition (gate cameras to perimeter cameras) with adversarial robustness against spoofing (masks, disguises, printed photos).

2. Identified Gaps

- Current Indian Army access control systems do not leverage AI-based facial or gait recognition for secondary or passive authentication.
- Long-range identification (100–500 m perimeter surveillance) using gait or silhouette analysis is not deployed.
- Spoofing robustness against printed photos, 3D masks and video replays is not addressed in existing biometric systems.
- Multi-sensor fusion (visible, near-infrared, thermal) for all-weather, all-lighting reliable identification is absent.
- Indigenous face/gait recognition models trained on Indian demographic diversity are not available.

3. Objectives

- Develop an integrated image-based identification system combining face, gait and body silhouette recognition.
- Implement robust liveness detection and anti-spoofing for presentation attacks.
- Build an Indian-demographic-representative training dataset ensuring equitable performance across diverse personnel.
- Integrate with existing access control infrastructure and alarm management systems.

4. Novelty in the Proposed Solution

The proposed system, PEHCHAAN (Perimeter-Enabled High-fidelity Composite Human Automated Access and Notification Network), introduces a Cascaded Multi-Modal Biometric Fusion (CMMBF) architecture that dynamically selects and weights identification modalities (face, gait, silhouette, IR signature) based on distance, lighting conditions and sensor availability. At long range where facial features are unresolvable, PEHCHAAN relies on gait rhythm analysis using a GCN-based skeleton estimator (MediaPipe + temporal GCN). A novel Demographic-Aware Training Protocol (DATP) constructs training batches with balanced representation across skin tone, gender and age groups from an indigenously collected dataset of Indian defence personnel (with consent). Anti-spoofing leverages depth maps and rPPG (remote photoplethysmography) to detect live presence.

5. Proposed Architecture

PEHCHAAN comprises: (1) Multi-Distance Camera Array (short-range 4K PTZ + medium-range thermal + long-range EO zoom); (2) Video Preprocessing Pipeline for super-resolution, deblurring and frame selection; (3) Face Recognition Module (AdaFace fine-tuned on Indian defence dataset, with FacePAD liveness detection); (4) Gait Recognition Module (skeleton extraction via MediaPipe, temporal GCN for gait embedding); (5) Silhouette Re-ID Module (SOLIDER backbone for body appearance); (6) CMMBF Fusion Layer combining modality scores using a learned attention weighting network conditioned on scene metadata; (7) Personnel Database with encrypted biometric templates using ISO/IEC 19794 standards; and (8) Access Control Integration Module interfacing with existing door control, CCTV, and alarm systems.

6. Conclusion

Securing military installations against infiltration, insider threats and impersonation requires identification systems that go beyond physical tokens to passive, continuous, and multi-modal biometric verification. PEHCHAAN will transform Indian Army access control by deploying AI-driven identification from the perimeter to the inner sanctum, providing layered security that is robust, equitable and fully indigenous — making unauthorised access exponentially more difficult while maintaining frictionless authentication for legitimate personnel.

10. Biometric-Based Multi-Factor Authentication for Secure Command Centers

1. Background

Command centres are the nerve centres of military operations, housing classified intelligence, C2 systems and nuclear command assets. Authentication to such facilities and systems must balance stringent security with operational speed. Multi-Factor Authentication (MFA) combining something you know, something you have and something you are (biometrics) has become the gold standard. The National Institute of Standards and Technology (NIST SP 800-63B) mandates biometric MFA for high-assurance federal systems. State-of-the-art biometric modalities include fingerprint (ISO/IEC 19794-4), iris recognition (ISO/IEC 19794-6), vein pattern recognition, voice recognition, and behavioural biometrics (keystroke dynamics, gait). Continuous authentication — re-verifying identity throughout a session rather than just at login — is an emerging paradigm demonstrated by DARPA's ACTIVE programme. Liveness detection standards (ISO/IEC 30107) address presentation attacks. In the Indian context, Aadhaar's Uidai biometric infrastructure demonstrates scale, but military-grade MFA for classified facilities requires offline capability, tamper-proof hardware security modules (HSM), and multi-modal fusion with extremely low false acceptance rates (FAR < 0.001%).

2. Identified Gaps

- Current Indian Army command centre access relies primarily on single-factor (card/PIN) or basic single-biometric authentication, insufficient for classified environments.
- Continuous authentication to detect post-authentication impersonation or coercion is not implemented.
- Offline, air-gapped biometric MFA systems with full capability independent of central servers are not deployed.
- Fusion of multiple biometric modalities to achieve FAR below 0.001% is not available as a sovereign solution.
- Coercion detection (identifying users operating under duress) is absent from existing authentication systems.

3. Objectives

- Develop a biometric MFA system fusing fingerprint, iris, face and voice modalities.
- Implement continuous authentication monitoring throughout command centre sessions using behavioural biometrics.
- Deploy a coercion detection module alerting silent alarms when users exhibit duress indicators.
- Ensure tamper-evident audit logging of all authentication events for forensic accountability.

4. Novelty in the Proposed Solution

The proposed system, SURAKSHA-MFA (Secure, Unified, Robust, and AI-driven Key-Controlled Holistic Authentication for Multi-Factor Access), introduces the concept of Adaptive Biometric Confidence Fusion (ABCF) where the system dynamically adjusts which biometric modalities are required based on the risk level of the action being authorised (e.g., viewing SECRET data vs authorising MOST SECRET communications). A pioneering Coercion Detection Layer analyses physiological signals (pupillary response, vocal stress, facial microexpressions) in real time during biometric capture to identify duress. Continuous authentication uses a Behavioural Biometric Shadow Model (BBSM) that learns individual typing cadence, mouse movement patterns and interaction dynamics, raising alarms on deviation. All biometric template storage and matching occurs within a DRDO-certified HSM, ensuring template privacy through homomorphic-encryption-assisted matching.

5. Proposed Architecture

SURAKSHA-MFA is structured as: (1) Biometric Capture Station with fingerprint (optical + capacitive), iris scanner (NIR 850 nm), face camera (visible + NIR), and microphone array; (2) Liveness Detection Engine (ISO 30107-3 compliant PAD for each modality); (3) ABCF Fusion Engine combining normalised match scores using a trained fusion neural network conditioned on access risk level; (4) Coercion Detection Module analysing pupillary dilation (IR camera), vocal tremor (audio analysis), and microexpression classifier; (5) Continuous Authentication Agent monitoring BBSM deviation throughout session; (6) HSM-Based Matching Engine performing encrypted template matching within certified hardware boundary; (7) Access Control Orchestrator granting, escalating, or revoking access based on continuous confidence scores; and (8) Tamper-Evident Audit Logger with blockchain-inspired chaining of log entries.

6. Conclusion

The security of command centres is the security of the nation. As authentication attacks grow more sophisticated — from social engineering and coercion to biometric spoofing — a layered, AI-powered multi-factor authentication system is not a luxury but a necessity. SURAKSHA-MFA provides the Indian Army with a sovereign, offline-capable, continuously authenticating and coercion-aware access control solution that meets the most stringent security requirements for classified command environments, ensuring that India's most critical decision-making infrastructure remains under the control of only the right people, at all times.